SMART HEALTH CARE MONITORING SYSTEM

Project Summary

Smart Health Care Monitoring System is an IoT-oriented healthcare monitoring web application. The deployed application, branded HealthGuard, provides patient vital entry, live sensor-status presentation, health dashboards, trend views, risk-analysis presentation, suggested precautions, patient records, and report-download access.

1. Project Objective

The project is designed to bring health-vital monitoring into a single web interface where patient information and vital measurements can be entered or represented from a connected ESP32 device, then viewed through monitoring, analysis, and record-management screens.

2. Core Features

- Patient vital input: name, age, gender, heart rate, SpO2, temperature and symptoms.
- ESP32 connected-device status is represented in the application.
- Live sensor feed for heart rate, SpO2 and temperature.
- Monitoring dashboard with heart-rate, SpO2 and temperature trends.
- Average temperature and latest-risk indicators.
- AI Risk Assessment interface with risk percentage.
- AI Suggested Precautions interface.
- Patient Records table with time, patient identity, vitals and risk level.
- Refresh control for patient records.
- Download Report control.
- System and sensor status indicators.

3. High-Level Architecture

The evidence-supported logical architecture is:

- Patient/User → Web Application
- ESP32 / sensing layer → health-vital data → application
- Web Application → Input Vitals
- Web Application → Dashboard / Trends
- Web Application → Risk Analysis presentation
- Web Application → Patient Records

The exact internal backend framework, database engine, API implementation, AI model/provider, sensor hardware and IoT communication protocol are not stated here because the actual Project 2 source code was not available in the supplied ZIP.

4. Functional Data Flow

1. A user opens the healthcare monitoring application.

2. The application displays system/device status and monitoring information.

3. Vital data can be entered through the Record Vital Signs interface.

4. The interface also provides a live sensor-feed area associated with the ESP32 device.

5. Heart rate, SpO2 and temperature are displayed and used in dashboard/trend views.

6. The analysis section presents risk assessment and suggested precautions when health data is available.

7. Patient information and vitals are presented in the records area.

8. The dashboard exposes a report-download action.

5. Frontend Modules

Home — Main entry/dashboard experience for the monitoring system.

Input Vitals — Captures patient identity and health-vital information.

Dashboard — Shows live/summary values, trends and risk-related indicators.

AI Analysis — Presents risk assessment and suggested precautions.

Records — Displays patient records and recorded vital information.

6. IoT / ESP32 Layer

The deployed interface explicitly identifies an ESP32 connected device and exposes a live sensor feed for heart rate, SpO2 and temperature. This establishes the IoT/device-monitoring component at the application level.

Not claimed without source verification: exact sensor modules, wiring/GPIO mapping, firmware implementation, sampling interval, payload structure and communication protocol.

7. Health Data

- Heart Rate
- SpO2
- Temperature
- Patient Name
- Age
- Gender
- Symptoms

The dashboard provides trend-oriented views for heart rate, SpO2 and temperature and also exposes an average-temperature indicator and latest-risk indicator.

8. Risk Analysis / AI Section

The application presents an 'AI Risk Assessment' area with a risk percentage and an 'AI Suggested Precautions' area. These are visible application capabilities.

The exact mechanism behind the analysis is not documented here as a specific ML model, rule engine, or external AI provider because that implementation cannot be verified from the currently supplied source.

9. Records & Reporting

- Patient Records view is provided.
- Records can be refreshed from the interface.
- Visible record fields include time, name, age/sex, heart rate, SpO2, temperature and risk level.
- A Download Report action is exposed on the dashboard.

10. Technologies

- IoT / ESP32-based healthcare monitoring concept.
- Web-based healthcare monitoring dashboard.
- Health-vital monitoring: heart rate, SpO2 and temperature.
- Patient data entry and records.
- Risk-analysis and precaution presentation.

15. Project Link

Live application: https://smart-health-monitoring-peach.vercel.app/